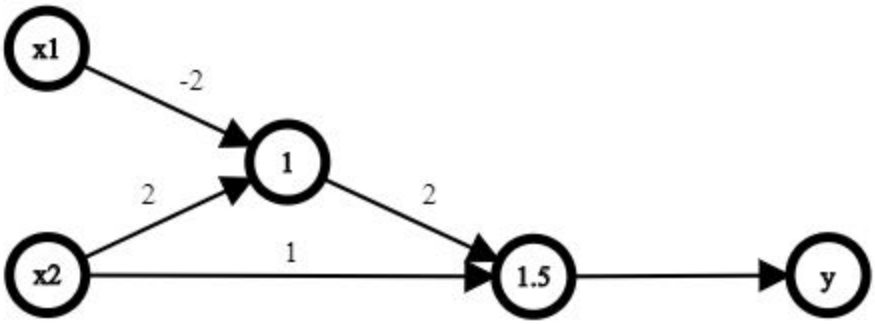
1. Which of the following boolean operations on two variables can be represented by a single perceptron?
   - **a. AND**
   - b. XOR
   - **c. OR**
   - **d. NOR**

   A single perceptron can represent any linearly separable function. AND, OR and NOR are linearly separable in 2D plane and hence a single perceptron can represent them.

2. Consider the following multilayer perceptron. The number on edge represents the weight of the input while the number inside the circle represents the threshold. x1 and x2 are boolean variables which can take values from {0,1}. What is the function represented by this perceptron network?



   a. x1 ∧ x2
   b. x1 ∨ x2
   c. x1 → x2
   d. **~x1 ∧ x2**

x1	x2	y
0	0	0
0	1	1
1	0	0
1	1	0

3. Consider a layer of neural network with the following weight matrix and bias vector.

$$W = \begin{pmatrix} 0.4 & 0.1 & 0.1 \\ -0.5 & 0.9 & 0.4 \\ -0.8 & 0.5 & 0.1 \\ -0.6 & 0.8 & -0.7 \end{pmatrix} b = \begin{pmatrix} 0.1 \\ -0.1 \\ -0.2 \\ 0.1 \end{pmatrix}$$

The input to this layer is

$$x = \begin{pmatrix} 0.4 \\ -1.0 \\ 0.1 \end{pmatrix}$$

If the activation function used is ReLU, what is the output of the layer?

a.

$$y = \begin{pmatrix} 0.17 \\ -1.16 \\ -1.01 \\ -1.01 \end{pmatrix}$$

b.

$$y = \begin{pmatrix} 0.17 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

c.

$$y = \begin{pmatrix} 0.07 \\ -1.06 \\ -0.81 \\ -1.11 \end{pmatrix}$$

d. $$y = \begin{pmatrix} 0.07 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

Wx = [0.07 -1.06 -0.81 -1.11]
Wx + b = [0.17 -1.16 -1.01 -1.01]
Relu(Wx + b) = [0.17 0 0 0]

4. What are the advantages of ReLU activation over tanh?
   a. **Faster to compute**
   b. **Reduces Gradient Vanishing**
   c. Zero centred output
   d. **Doesn't saturate at high values**

RelU is faster to compute since it requires just a comparison to zero instead of transcendental function evaluation in case of tanh. ReLU maintains a gradient of 1 everywhere above 0 thus doesn't suffer gradient vanishing. It doesn't saturate whereas tanh saturates at values far away from 0.

5. How many times does the size of the input decrease after passing through a 3x3 MaxPool layer with stride 3? Size of the input is defined as height x breadth x depth.
   Ans. 9

Pooling will reduce height and breadth by 3 times each whereas depth will stay the same. So the size reduces by 9 times.

6. Consider a neural network that performs the following operations:
   $h = Wx + b$
   $y = \tanh(h)$,
   $J = \Sigma_i d_i y_i$
   where $x \in \mathbb{R}^{n \times 1}$, $W \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^{m \times 1}$, $d \in \mathbb{R}^{m \times 1}$.
   Which of these represents the derivative $\partial J / \partial b$?
   a. **$d \circledast (1 - y \circledast y)$**
   b. $d \circledast (1 - h \circledast h)^T$
   c. $d.(1 - y \circledast y)$
   d. $d.(1 - h \circledast h)$

$\partial J / \partial y_i = d_i$
$\partial y_i / \partial h_i = (1 - y_i \,.\, y_i)$
$\partial h_i / \partial b_i = 1$

$\partial J / \partial b = [\partial J / \partial b_i]_{\ i = 1 \text{ to } m}$
$= [\partial J / \partial h_i \,.\, \partial h_i / \partial b_i]_{\ i = 1 \text{ to } m}$
$= [\partial J / \partial y_i \,.\, \partial y_i / \partial h_i \,.\, \partial h_i / \partial b_i]_{\ i = 1 \text{ to } m}$
$= [d_i \,.\, (1 - y_i \,.y_i)]_{\ i = 1 \text{ to } m}$
$= d \circledast (1 - y \circledast y)$

Hence, option a is correct.

7. Consider a neural network that performs the following operations:
   $h = Wx + b$
   $y = \log(h)$,
   $J = \Sigma_i \, y_i^2$
   where $x \in \mathbb{R}^{n \times 1}$, $W \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^{m \times 1}$.
   Which of the represents the derivative $\partial J / \partial b$?
   a. h./y
   b. 2y./y
   c. 2log(h)./h
   d. 2h./log(h)

   $\partial J / \partial y_i = 2y_i$
   $\partial y_i / \partial h_i = 1/h_i$
   $\partial h_i / \partial b_i = 1$

   $\partial J / \partial b = [\partial J / \partial b_i]_{\ i = 1 \text{ to } m}$
   $= [\partial J / \partial h_i \,.\, \partial h_i / \partial b_i]_{\ i = 1 \text{ to } m}$
   $= [\partial J / \partial y_i \,.\, \partial y_i / \partial h_i \,.\, \partial h_i / \partial b_i]_{\ i = 1 \text{ to } m}$
   $= [2y_i / h_i]_{\ i = 1 \text{ to } m}$
   $= 2y./h$
   $= 2\log(h)./h$

   Hence, option c is correct.

8. The tanh activation performs better than sigmoid activation function for hidden units in terms of accuracy for a particular task. What might be the reason for this?

a. The tanh function does not suffer from the vanishing gradient problem but sigmoid function does.
b. Derivative computation of tanh is performance efficient compared to the derivative computation of sigmoid.
c. **None of the above**

Sigmoid and Tanh both suffer from the vanishing gradient problem. This problem is solved by using the ReLu activation function therefore option A is incorrect. Performance efficiency for computing the derivative of tanh is no reason for it to perform better than the sigmoid function. A neural model that trains faster does not imply it has to perform better therefore option B is incorrect.

9. What is the size of the output image after an input image of size 64 x 64 is passed through a convolution filter of size 4 x 4 and stride 2 followed by a convolution filter of size 7 x 7 and stride 1?
   a. 26 x 26
   b. 25 x 25
   c. 24 x 24
   d. 23 x 23

   Input size: 64 x 64
   After passing through the first layer: (64 - 4) / 2 + 1 = 31 i.e. 31 x 31
   After passing through the second layer: (31 - 7) + 1 = 25 i.e. 25 x 25
   Hence, option b is correct.

10. Statement 1: Initializing all parameters of a neural network to zero helps learning.
    Statement 2: Using max pooling in CNN, increases the no. of parameters.

    Which of the following is correct?

    a. Both the statements are correct.
    b. Both the statements are incorrect.
    c. Statement 1 is correct and Statement 2 is incorrect.
    d. Statement 2 is correct and Statement 1 is incorrect.

    If all the parameters of a neural network are initialized to zero then every node in a layer is basically computing the same value. Even after backpropagation, the weights are changed by the same value. Such a neural network is equivalent to a network with only a single node in every layer which impairs the performance therefore Statement 1 is incorrect. Max pooling layer does not involve any parameters. It only computes the max over a set of blocks in the input image therefore Statement 2 is incorrect.